COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	10/5/26	**DATE SUBMITTED:**	10/6/26
NAME:	Villareal, Vanderohe Vernon	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 10 **Install, Configure and Manage Enterprise Log Monitoring via Ansible**

1. PROCEDURE OUTPUT

Step 1:
Create roles using mkdir command

```
villareal@villareal-ansible1:~/HOA10$ mkdir -p roles/elasticsearch/tasks
villareal@villareal-ansible1:~/HOA10$ mkdir -p roles/elasticsearch/handlers
villareal@villareal-ansible1:~/HOA10$ mkdir -p roles/kibana/tasks
villareal@villareal-ansible1:~/HOA10$ mkdir -p roles/kibana/handlers
villareal@villareal-ansible1:~/HOA10$ mkdir -p roles/logstash/handlers
villareal@villareal-ansible1:~/HOA10$ mkdir -p roles/logstash/tasks
```

```
villareal@villareal-ansible1:~/HOA10$ tree
.
├── inventory
└── roles
    ├── elasticsearch
    │   ├── handlers
    │   └── tasks
    ├── kibana
    │   ├── handlers
    │   └── tasks
    └── logstash
        ├── handlers
        └── tasks
```

Step 2:
Create an inventory file

```
  GNU nano 7.2                                  inventory
[elasticsearch]
192.168.56.109

[kibana]
192.168.56.109

[logstash]
192.168.56.110
```

Step 3:
Create roles for each (elasticsearch, kibana, and logstash)

Elasticsearch

GNU nano 7.2 roles/elasticsearch/tasks/main.yml

```yaml
---
- name: Install required packages
  apt:
    name:
      - apt-transport-https
      - ca-certificates
      - wget
      - gnupg
    state: present
    update_cache: yes

- name: Download Elasticsearch GPG key
  get_url:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    dest: /tmp/GPG-KEY-elasticsearch

- name: Add Elasticsearch GPG key
  apt_key:
    file: /tmp/GPG-KEY-elasticsearch
    state: present

- name: Add Elasticsearch repository
  apt_repository:
    repo: "deb https://artifacts.elastic.co/packages/8.x/apt stable main"
    state: present
    filename: elastic-8.x
```

```yaml
- name: Install Elasticsearch
  apt:
    name: elasticsearch
    state: present
    update_cache: yes

- name: Enable and start Elasticsearch
  systemd:
    name: elasticsearch
    enabled: yes
    state: started
```

Kibana

```
  GNU nano 7.2                    roles/kibana/tasks/main.yml
---
- name: Install required packages
  apt:
    name:
      - apt-transport-https
      - ca-certificates
      - wget
      - gnupg
    state: present
    update_cache: yes

- name: Download Elasticsearch GPG key
  get_url:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    dest: /tmp/GPG-KEY-elasticsearch

- name: Add Elasticsearch GPG key
  apt_key:
    file: /tmp/GPG-KEY-elasticsearch
    state: present

- name: Add Elastic repository
  apt_repository:
    repo: "deb https://artifacts.elastic.co/packages/8.x/apt stable main"
    state: present
    filename: elastic-8.x
```

```
- name: Install Kibana
  apt:
    name: kibana
    state: present
    update_cache: yes

- name: Enable and Start Kibana
  systemd:
    name: kibana
    enabled: yes
    state: started
```

Logstash

```
  GNU nano 7.2                    roles/logstash/tasks/main.yml
---
- name: Install required packages
  yum:
    name:
      - java-11-openjdk
      - wget
    state: present

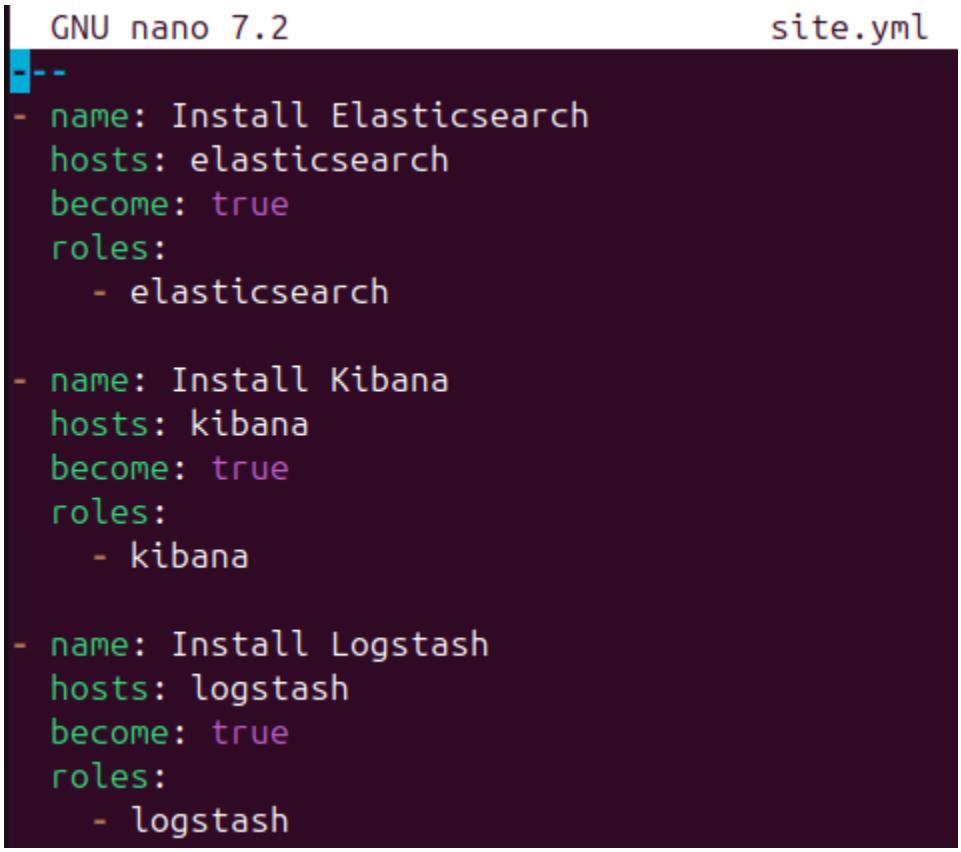
- name: Download Elastic repository
  get_url:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    dest: /etc/pki/rpm-gpg/GPG-KEY-elasticsearch

- name: Add Elastic repository
  yum_repository:
    name: elastic-8.x
    description: Elastic repository for 8.x packages
    baseurl: https://artifacts.elastic.co/packages/8.x/yum
    gpgcheck: yes
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: yes

- name: Install Logstash
  yum:
    name: logstash
    state: present
```

```
- name: Enable and start Logstash
  systemd:
    name: logstash
    enabled: yes
    state: started
```

Step 4:
Create the main playbook



Step 5:
Run the playbook

```
villareal@villareal-ansible1:~/HOA10$ ansible-playbook -i inventory site.yml --a
sk-become-pass
BECOME password:

PLAY [Install Elasticsearch] ***************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.109]

TASK [elasticsearch : Install required packages] *******************************
changed: [192.168.56.109]

TASK [elasticsearch : Download Elasticsearch GPG key] **************************
changed: [192.168.56.109]

TASK [elasticsearch : Add Elasticsearch GPG key] *******************************
changed: [192.168.56.109]

TASK [elasticsearch : Add Elasticsearch repository] ****************************
changed: [192.168.56.109]

TASK [elasticsearch : Install Elasticsearch] ***********************************
changed: [192.168.56.109]

TASK [elasticsearch : Enable and start Elasticsearch] **************************
changed: [192.168.56.109]
```

```
PLAY [Install Kibana] **********************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.109]

TASK [kibana : Install required packages] **************************************
ok: [192.168.56.109]

TASK [kibana : Download Elasticsearch GPG key] *********************************
ok: [192.168.56.109]

TASK [kibana : Add Elasticsearch GPG key] **************************************
ok: [192.168.56.109]

TASK [kibana : Add Elastic repository] *****************************************
ok: [192.168.56.109]

TASK [kibana : Install Kibana] *************************************************
ok: [192.168.56.109]

TASK [kibana : Enable and Start Kibana] ****************************************
changed: [192.168.56.109]
```

```
PLAY [Install Logstash] ****************************************************

TASK [Gathering Facts] *****************************************************
ok: [192.168.56.110]

TASK [logstash : Install required packages] ********************************
changed: [192.168.56.110]

TASK [logstash : Download Elastic repository] ******************************
changed: [192.168.56.110]

TASK [logstash : Add Elastic repository] ***********************************
changed: [192.168.56.110]

TASK [logstash : Install Logstash] *****************************************
changed: [192.168.56.110]

TASK [logstash : Enable and start Logstash] ********************************
changed: [192.168.56.110]

PLAY RECAP *****************************************************************
192.168.56.109             : ok=14   changed=8    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
192.168.56.110             : ok=6    changed=5    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
```

Step 6:
Verify the installation using the commands

Elasticsearch on Ubuntu

```
villareal@villareal-ansible1:~/HOA10$ ansible elasticsearch -i inventory -m she
l -a "systemctl status elasticsearch --no-pager"
192.168.56.109 | CHANGED | rc=0 >>
● elasticsearch.service - Elasticsearch
     Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; enabled; pr
set: enabled)
     Active: active (running) since Tue 2026-10-06 10:43:54 UTC; 20min ago
       Docs: https://www.elastic.co
```

Kibana on Ubuntu

```
villareal@villareal-ansible1:~/HOA10$ ansible kibana -i inventory -m shell -a "s
ystemctl status kibana --no-pager"
192.168.56.109 | CHANGED | rc=0 >>
● kibana.service - Kibana
     Loaded: loaded (/usr/lib/systemd/system/kibana.service; enabled; preset: en
abled)
     Active: active (running) since Tue 2026-10-06 11:07:33 UTC; 1min 4s ago
       Docs: https://www.elastic.co
   Main PID: 9523 (MainThread)
      Tasks: 11 (limit: 3447)
     Memory: 629.0M (peak: 641.8M)
        CPU: 44.309s
     CGroup: /system.slice/kibana.service
             └─9523 /usr/share/kibana/bin/../node/glibc-217/bin/node /usr/share/
kibana/bin/../node_modules/@kbn/cli/kibana/dist
```

Logstash on CentOS

```
villareal@villareal-ansible1:~/HOA10$ ansible logstash -i inventory -m shell -a
"systemctl status logstash --no-pager"
192.168.56.110 | CHANGED | rc=0 >>
● logstash.service - logstash
     Loaded: loaded (/usr/lib/systemd/system/logstash.service; enabled; preset:
disabled)
     Active: active (running) since Tue 2026-10-06 19:04:38 PST; 8s ago
   Main PID: 59612 (java)
      Tasks: 23 (limit: 22978)
     Memory: 229.8M (peak: 234.5M)
        CPU: 16.331s
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:
Answer the following:
1. What are the benefits of having log monitoring tool?
   - Log monitoring tools help detect system problems, security issues, and important events. They also help administrators analyze logs, find errors, and prevent downtime.

Conclusions:
In this activity, we learned how to install, configure, and manage a log monitoring tool using Ansible. We used the Elastic Stack to help collect, monitor, and analyze system logs. This activity also showed how Ansible roles can make the installation more organized and easier to manage. Overall, log monitoring is useful for finding system errors, detecting possible security problems, and maintaining the performance and reliability of IT systems.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

COMPUTER ENGINEERING

QUEZON CITY

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Error encountered and explanation of new commands for the activity**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI is used to help resolve issues encountered during pushing of the activity to a fresh GitHub and**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.